

Coding & Solution

Date	15 March 2026
Team ID	xxxxxx
Project Name	AI based crop recommendation
Maximum Marks	5 Marks

Coding & Solution

This section presents the implementation of the OptiCrop – AI-Based Crop Recommendation System. The solution uses machine learning to recommend the most suitable crop based on soil nutrients (Nitrogen, Phosphorus, Potassium), temperature, humidity, pH, and rainfall. The application is developed using Python and Flask, following clean coding practices, modular design, and project requirements to ensure reliability, maintainability, and ease of future enhancements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack (Python, Flask, HTML, CSS, and Machine Learning libraries).
2. The solution should address the core problem statement by recommending the most suitable crop based on soil nutrients and environmental conditions.
3. Include all source files, trained model files, static assets, templates, and configuration files required to run the application.
4. Attach or link to your GitHub repository containing the complete project code and documentation.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/Sandhali6/Opti_Crop
Programming Language(s)	Python, HTML, CSS
Framework(s) Used	Flask, Scikit-learn, Pandas, NumPy
Key Features Implemented	Crop recommendation using Machine Learning, data preprocessing, trained ML model, responsive web interface, soil parameter input, crop prediction display

Pending / Incomplete Features	User authentication, crop information database, fertilizer recommendation, weather API integration, multilingual support
Setup / Run Instructions	Install dependencies using <code>pip install -r requirements.txt</code> , then run <code>python app.py</code> and open the application in a web browser.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The OptiCrop project successfully integrates a Machine Learning model with a Flask-based web application to recommend the most suitable crop for cultivation based on soil and environmental parameters. The application follows a modular architecture, making the code easy to understand, maintain, and extend. Future enhancements such as fertilizer recommendation, real-time weather integration, multilingual support, and user authentication can further improve the usability and effectiveness of the system.